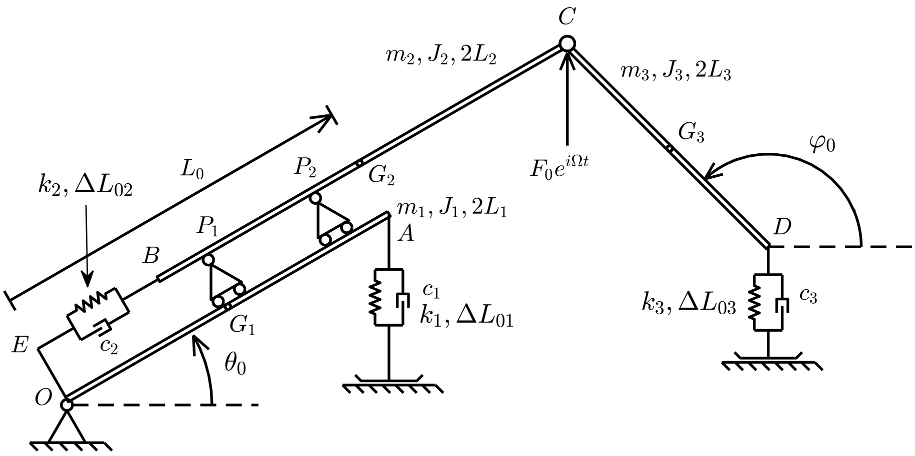
Dynamics of Mechanical Systems

June 09th, 2022

Problem 1:



$m_1 =$ 10 kg	$\Delta L_{01} =$ -0.0127 m
$m_2 =$ 10 kg	$\Delta L_{02} =$ -0.0058 m
$m_3 =$ 5 kg	$\Delta L_{03} =$ -0.0016 m
$J_1 =$ 0.8 kgm^2	$c_1 =$ 2.0 Ns/m
$J_2 =$ 2.0 kgm^2	$c_2 =$ 2.0 Ns/m
$J_3 =$ 0.15 kgm^2	$c_3 =$ 0.5 Ns/m
$L_1 =$ 0.5 m	$\theta_0 = \pi/6$
$L_2 =$ 0.75 m	$\varphi_0 = {}^{2}/_{3}\,\pi$
$L_3 =$ 0.3 m	$\overline{BP_1} = 0.1\ m$
$L_0 =$ 1.0 m	$\overline{P_1P_2} = 0.3\ m$
$k_1 =$ 15000 N/m	
$k_2 =$ 15000 N/m	
$k_3 =$ 15000 N/m	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The first rigid bar OA is connected to the ground through a hinge O; the second rigid bar BC can slide with respect to the first one and is connected to the OE side through a spring and damper system (k_2, c_2) – consider zero the distance OE and, accordingly, the dimensions of the two carts P_2 and P_1 that allow for the relative displacements. A third rigid bar is connected to the second through the hinge C. In the equilibrium position the distance EG_2 is L_0.

The students are requested to:

0. Write on top of each page their surname, name, Person Code and **SNxxxx code** (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration (provide this answer on paper: Jacobian matrices and second-order stiffness terms)
2. Compute the natural frequencies and modal shapes (provide this answer on paper, copying the values from MATLAB);
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic vertical force $F_0 e^{i\Omega t}$. Output: the vertical component of the displacement of point C and the horizontal component of the displacement of point D. Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with SNxxxx your identifier, see below;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic vertical force $F_0 e^{i\Omega t}$. Output: the horizontal component of the constraint force in the hinge O. Save the graphical result as SNxxxx3.fig.
5. Compute the static preloads ΔL_{01}, ΔL_{02} and ΔL_{03} when $\theta_0 = \pi/4$ and $\varphi_0 = \pi$ (provide the answer on paper).

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

You must upload one single archive file for both problem 1 and problem 2, which should contain your MATLAB scripts and all the .fig files you want to deliver as part of your solution.

i.) Accepted formats for the archive files are: .7z .rar .zip;
ii.) The file must be named SNxxxx with "S" the first letter in your surname, "N" the first letter in your name (in case of multiple surnames/names just consider the first one), "xxxx" the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
iii.) The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test_09_Jun_2022**;
iv.) the deadline to upload the assignment will be announced during the written exam.

Your identifier: SNxxxx with S the first letter of your surname, N the first letter of your name and xxxx the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the identifier BS5678

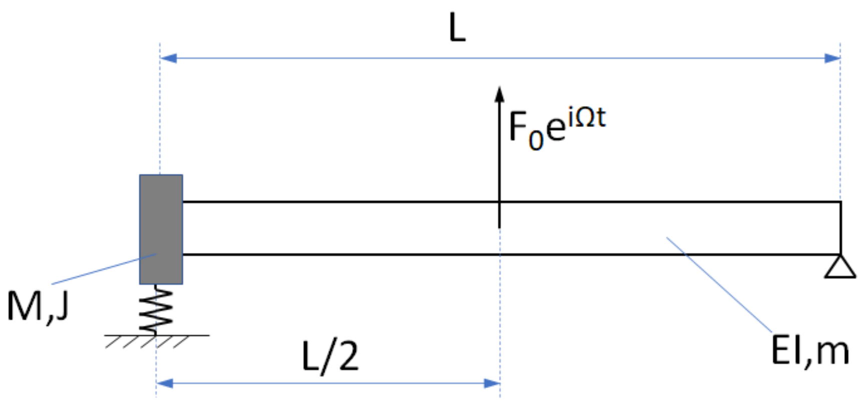
Please be aware that your answers to the test will not be considered in case:

a. **the information stated at point 0 above is missing;**
b. **your archive file is uploaded in the wrong folder**
c. **the name of the archive file is different from the identifier SNxxxx described above;**

For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one.

Problem 2:

Consider the <u>bending motion</u> of a beam with uniform section and material properties having length L. The left end of the beam is rigidly attached to a rigid body having mass M and moment of inertia J with respect to its centre of gravity. A linear spring having stiffness k is connected to the rigid body at its upper end, while the lower end is fixed to the ground. The right end of the beam is constrained by a hinge/pin.

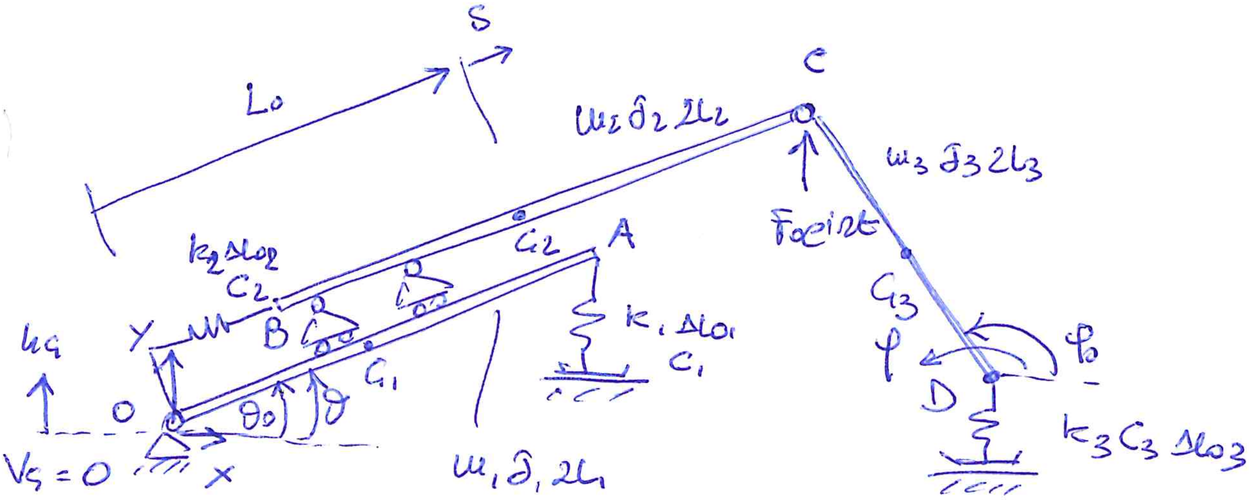


DATA:
$EI=4e5\ Nm^2$
$m=10\ kg/m$
$L=2\ m$
$k=1e7\ N/m$
$M=10kg$
$J=0.1\ kgm^2$

1) Write the boundary conditions and the expression of the square matrix of coefficients H of the linear homogeneous system defining the natural frequencies of the system. Deliver this part of the solution on paper.
2) Compute the natural frequencies of the system in the frequency range 0-200 Hz and the related modal shapes. Deliver the values of the natural frequencies on paper and include in your archive file one .fig file for each modal shape. The .fig files shall be named SNxxxxP2i with SNxxxx your identifier (see below) and i=1,2,3 (use as many as needed);
3) Plot the Bode diagram (log y scale) of the magnitude of the FRF considering:
   - Non-dimensional damping factor: 0.02 for all modes
   - input: force having unit magnitude applied in the centre of beam as shown in the figure;
   - output: vertical vibration at the left end of the beam;

   The Bode diagram shall be delivered as a .fig file named SNxxxxP2FRF included in your archive.

WRITTEN TEST - PROBLEM 1



3 rigid bodies → 9 dof without constraints

2 hinges O and C → −4 dof → 3 dof

2 carts between m_1 and m_2 → −2 dof

$$\underline{x} = \begin{Bmatrix} \theta \\ s \\ \varphi \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ 0 \\ \varphi_0 \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 V_{G1}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} m_2 \dot{y}_{G2}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_{G3}^2 + \frac{1}{2} m_3 \dot{y}_{G3}^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

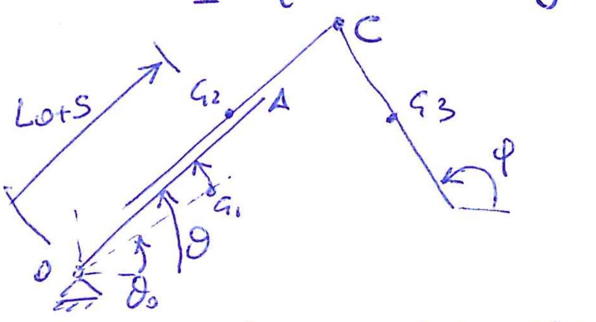
$$D = \frac{1}{2} \sum_{i=1}^{3} c_i \Delta \dot{l}_i^2 \ ; \quad V_{el} = \frac{1}{2} \sum_{i=1}^{3} k_i \Delta l_i^2 \ ; \quad V_g = \sum_{i=1}^{3} m_i g h_{Gi}$$

$$\delta W = F_0 e^{i\Omega t} \delta y_C$$

$$[M_{PH}] = diag(m_1\ J_1\ m_2\ m_2\ J_2\ m_3\ m_3\ J_3) \ ; \ \underline{y}_M = \{s_{G1}\ \theta\ x_{G2}\ y_{G2}\ \theta\ x_{G3}\ y_{G3}\ \varphi\}^T$$

$$[C_{PH}] = diag(c_1\ c_2\ c_3) \ ; \ [K_{PH}] = diag(k_1\ k_2\ k_3) \ ; \ \underline{y}_c = \underline{y}_k = \{\Delta l_1\ \Delta l_2\ \Delta l_3\}^T$$

$$\underline{F} = \{F_0 e^{i\Omega t}\} \ ; \ y_F = \{y_C\}$$



• non linear kinematic relationships:

$$V_{G1} = L_1 \dot{\theta} \rightarrow S_{G1} = L_1(\theta - \theta_0)$$

$y_A = 2L_1 \sin\theta$

$x_{G2} = (L_0 + S)\cos\theta$

$y_{G2} = (L_0 + S)\sin\theta$

$x_C = (L_0 + L_2 + S)\cos\theta$

$y_C = (L_0 + L_2 + S)\sin\theta$

- the trajectory of G_1 is a circle with radius $= L_1$
- the two carts allow for a relative translation along the direction $\overline{OA}$ → → the relative rotation is zero

$$x_{s3} = (L_0 + L_2 + S)\cos\vartheta - L_3 \cos\varphi$$
$$y_{s3} = (L_0 + L_2 + S)\,\text{sen}\,\vartheta - L_3\,\text{sen}\,\varphi$$
$$y_D = (L_0 + L_2 + S)\,\text{sen}\,\vartheta - 2L_3\,\text{sen}\,\varphi$$

$$\Delta L_1 = y_A - y_{A\phi} = 2L_1\,\text{sen}\,\vartheta - y_{A\phi}$$
$$\Delta L_2 = S$$
$$\Delta L_3 = y_D - y_{D\phi} = (L_0 + L_2 + S)\,\text{sen}\,\vartheta + \\ - 2L_3\,\text{sen}\,\varphi - y_{D\phi}$$

$$\frac{\partial x_{s2}}{\partial \vartheta} = (L_0 + S)(-\text{sen}\,\vartheta) \qquad \frac{\partial x_{s2}}{\partial S} = \cos\vartheta$$

$$\frac{\partial y_{s2}}{\partial \vartheta} = (L_0 + S)\cos\vartheta \qquad \frac{\partial y_{s2}}{\partial S} = \text{sen}\,\vartheta$$

$$\frac{\partial x_{s3}}{\partial \vartheta} = (L_0 + L_2 + S)(-\text{sen}\,\vartheta) \qquad \frac{\partial x_{s3}}{\partial S} = \cos\vartheta \qquad \frac{\partial x_{s3}}{\partial \varphi}\; L_3\,\text{sen}\,\varphi$$

$$\frac{\partial y_{s3}}{\partial \vartheta} = (L_0 + L_2 + S)\cos\vartheta \qquad \frac{\partial y_{s3}}{\partial S} = \text{sen}\,\vartheta \qquad \frac{\partial y_{s3}}{\partial \varphi} = -L_3\cos\varphi$$

$$[\Lambda_M] = \begin{bmatrix} L_1 & 0 & 0 \\ 1 & 0 & 0 \\ -L_0\,\text{sen}\,\vartheta_0 & \cos\vartheta_0 & 0 \\ L_0\cos\vartheta_0 & \text{sen}\,\vartheta_0 & 0 \\ 1 & 0 & 0 \\ -(L_0+L_2)\,\text{sen}\,\vartheta_0 & \cos\vartheta_0 & L_3\,\text{sen}\,\varphi_0 \\ (L_0+L_2)\cos\vartheta_0 & \text{sen}\,\vartheta_0 & -L_3\cos\varphi_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[\Lambda_k] = [\Lambda_c] = \begin{bmatrix} 2L_1\cos\vartheta_0 & 0 & 0 \\ 0 & 1 & 0 \\ (L_0+L_2)\cos\vartheta_0 & \text{sen}\,\vartheta_0 & -2L_3\cos\varphi_0 \end{bmatrix}$$

$$[\Lambda_0] = \begin{bmatrix} (L_0+L_2)\cos\vartheta_0 & \text{sen}\,\vartheta_0 & 0 \end{bmatrix}$$

$$\begin{cases} u_{s1} = y_{s1} \\ u_{s2} = y_{s2} \\ u_{s3} = y_{s3} \end{cases} \rightarrow [k_{s1}] = m_1 g \left[\frac{\partial^2 u_{s1}}{\partial x_i x_j}\right] = m_1 g \begin{bmatrix} -L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[k_{s2}] = m_2 g \begin{bmatrix} -L_0 \operatorname{sen}\theta_0 & \cos\theta_0 & 0 \\ \cos\theta_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [k_{s3}] = m_3 g \begin{bmatrix} -(L_0+L_2)\operatorname{sen}\theta_0 & \cos\theta_0 & 0 \\ \cos\theta_0 & 0 & 0 \\ 0 & 0 & L_3 \operatorname{sen}\varphi_0 \end{bmatrix}$$

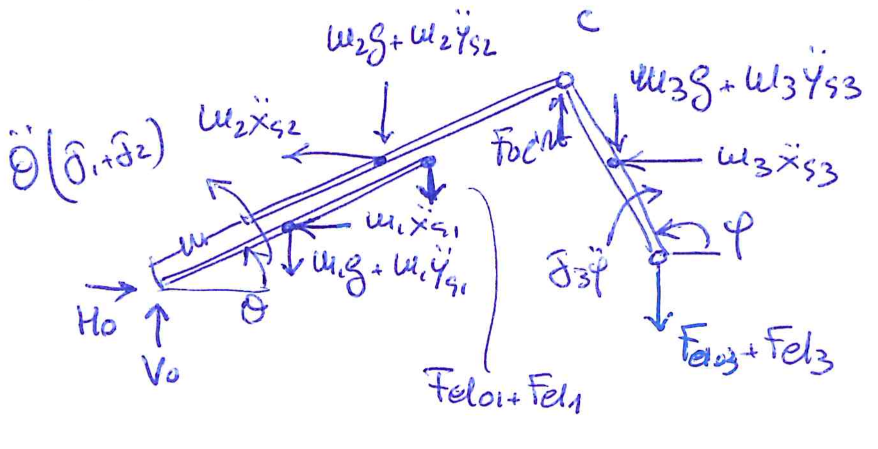
$$[k_{el_{II}1}] = k_1 \Delta l_{01} \left[\frac{\partial^2 \Delta l_1}{\partial x_i \partial x_j}\right] = k_1 \Delta l_{01} \begin{bmatrix} -2L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[k_{el_{II}2}] = \emptyset \quad [k_{el_{II}3}] = k_3 \Delta l_{03} \begin{bmatrix} -(L_0+L_2)\operatorname{sen}\theta_0 & \cos\theta_0 & 0 \\ \cos\theta_0 & 0 & 0 \\ 0 & 0 & 2L_3 \operatorname{sen}\varphi_0 \end{bmatrix}$$

- question 3:

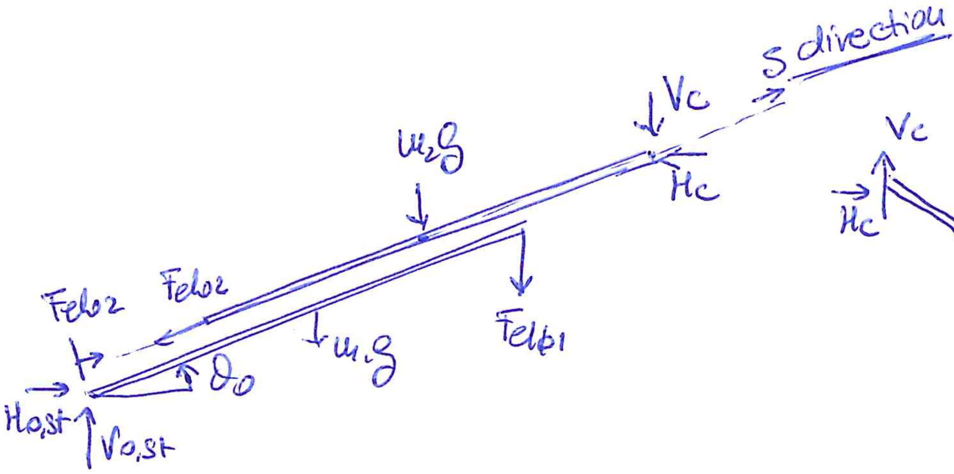
$$\left.\begin{aligned} y_{C,lin} &= \left.\frac{\partial y_C}{\partial \theta}\right|_{\underline{x}_0} (\theta-\theta_0) + \left.\frac{\partial y_C}{\partial s}\right|_{\underline{x}_0} s + \left.\frac{\partial y_C}{\partial \varphi}\right|_{\underline{x}_0} (\varphi-\varphi_0) \\ x_{D,lin} &= \left.\frac{\partial x_D}{\partial \theta}\right|_{\underline{x}_0} (\theta-\theta_0) + \left.\frac{\partial x_D}{\partial s}\right|_{\underline{x}_0} s + \left.\frac{\partial x_D}{\partial \varphi}\right|_{\underline{x}_0} (\varphi-\varphi_0) \end{aligned}\right\} \text{linearized outputs}$$

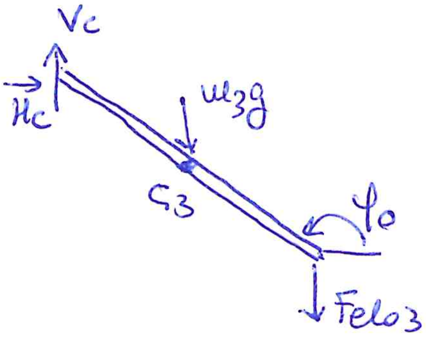
- question 4



$$\Sigma F_x = 0 \rightarrow H_0 = m_1 \ddot{x}_{s1} + m_2 \ddot{x}_{s2} + m_3 \ddot{x}_{s3}$$

$$\ddot{x}_{si} = -\Omega^2 x_{si,lin}$$





$$\sum M_C^{m_3} = 0 \quad : \quad \left(m_3 g L\cancel{/3} + F_{elo3}\, 2L\cancel{/3}\right)\cancel{\cos\varphi_0} = 0 \quad \rightarrow \quad F_{elo3} = -\frac{m_3 g}{2}$$

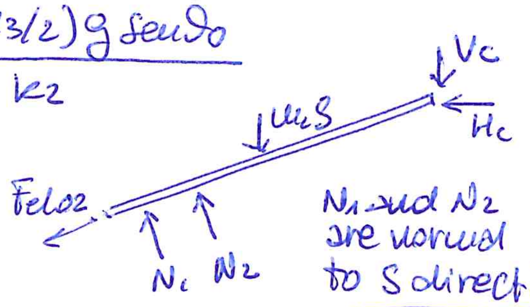
$$\Delta L_{03} = -\frac{m_3 g}{2k_3}$$

$$\sum F_y^{m_3} = 0 \quad : \quad V_C - m_3 g - F_{elo3} = 0$$

$$\hookrightarrow V_C = \frac{m_3 g}{2}$$

$$\sum F^{m_2}_{s\_direction} = 0 \quad : \quad F_{elo2} + m_2 g \,\mathrm{sen}\,\vartheta_0 + V_C \,\mathrm{sen}\,\vartheta_0 = 0$$

$$\Delta L_{02} = -\frac{(m_2 + m_3/2)\, g \,\mathrm{sen}\,\vartheta_0}{k_2}$$



N_1 and N_2 are normal to S direction

$$\sum M_O^{m_1+m_2} = 0 : m_1 g L_1 \cancel{\cos\vartheta_0} + F_{elo1}\, 2L_1 \cancel{\cos\vartheta_0} + m_2 g L_0 \cancel{\cos\vartheta_0} + V_C (L_0 + L_2) \cancel{\cos\vartheta_0} = 0$$

$$\Delta L_{01} = -\frac{m_1 g L_1 + m_2 g L_0 + m_3/2\, g\, (L_0 + L_2)}{2 k_1 L_1}$$